

Health+ Diet Planner

A Smart Health & Nutrition Management Platform

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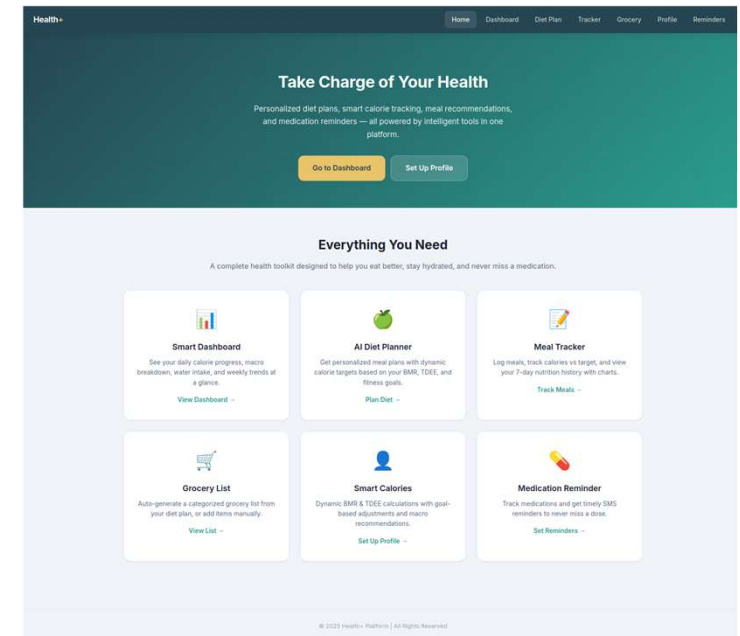
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1. Introduction & Overview

What is Health+ Diet Planner?

A comprehensive web-based health management platform
Combines personalized diet planning with smart calorie tracking
Integrates AI-powered meal recommendations via Google Gemini API
Provides automated grocery list generation from meal plans
Includes water intake monitoring with daily goal tracking
Supports medication reminder system via Twilio SMS
All user data stored locally in browser (localStorage) for privacy
Mobile-responsive design using modern CSS custom properties



2. Problem Statement

Challenges in Existing Systems

- Static calorie goals — no personalization
- No integrated meal tracking with history
- Manual grocery planning is time-consuming
- Scattered tools — no single platform solution
- Poor mobile responsiveness in existing apps

Our Solution: Health+

- Dynamic BMR + TDEE calorie calculation
- 7-day meal history with nutritional analytics
- Auto-generate grocery lists from diet plans
- All-in-one platform: diet, tracking, reminders
- Mobile-first responsive design

Project Objectives

- Provide personalized calorie targets based on user biometrics and fitness goals
- Enable daily meal logging with real-time nutritional progress tracking
- Generate AI-powered diet plans using Google Gemini API integration
- Automate grocery list creation from meal plans to save user time

3. System Architecture

Presentation Layer

HTML5, CSS3, Chart.js, Responsive UI



Business Logic Layer

Vanilla JavaScript Modules (calories.js, storage.js, recommendations.js, ui.js)



Data Storage Layer

localStorage (JSON-serialized) — Profile, Meals, Water, Diet Plan, Grocery



API Integration Layer

Express.js Backend → Google Gemini API + Twilio SMS API

4. Key Features — 8 Modules



Smart Calorie System

BMR + TDEE calculation
Goal-based adjustment



User Profile

Weight, height, age, gender
localStorage persistence



Dashboard

Stat cards, Chart.js charts
Weekly trends + macros



Meal Tracker

Log meals, consumed vs target
7-day history



Meal Recommendations

50+ meals filtered by
remaining calories/protein



Grocery Generator

Auto-generate from diet plan
Manual items + categories



Water Tracker

±1 glass counter
Daily goal progress bar



AI Diet Plans

Google Gemini API
Personalized meal plans

5. Technology Stack

Frontend

HTML5 — Semantic markup, accessibility
CSS3 — Custom properties, Flexbox/Grid, media queries
JavaScript (ES6+) — Modular architecture, async/await
Chart.js v4.4 — Doughnut & bar charts for analytics

Backend

Node.js v18+ — Server runtime
Express.js v4.21 — REST API framework
Google Gemini 1.5 Flash — AI diet plan generation
Twilio SDK — SMS medication reminders

Storage & Tools

localStorage — Client-side JSON persistence
dotenv — Environment variable management
CORS — Cross-origin resource sharing
Git/GitHub — Version control & collaboration

6. Smart Calorie System

Step 1: Calculate BMR (Mifflin-St Jeor)

Male: $BMR = 10 \times \text{weight} + 6.25 \times \text{height} - 5 \times \text{age} + 5$

Female: $BMR = 10 \times \text{weight} + 6.25 \times \text{height} - 5 \times \text{age} - 161$

Step 2: Calculate TDEE

$TDEE = BMR \times \text{Activity Multiplier}$

Sedentary: $\times 1.2$ | Light: $\times 1.375$ | Moderate: $\times 1.55$

Active: $\times 1.725$ | Very Active: $\times 1.9$

Step 3: Goal-Based Adjustment

Weight Loss: $TDEE - 500 \text{ kcal}$

Muscle Gain: $TDEE + 300 \text{ kcal}$

Maintenance: TDEE (no change)

Safety floor: minimum 1200 kcal/day

The screenshot shows a web application interface for a health and nutrition tracker. The header includes a navigation bar with links: Home, Dashboard, Diet Plan, Tracker, Grocery, Profile (active), and Reminders. The main section is titled 'My Profile' with a subtitle 'Set your personal details for accurate calorie and nutrition calculations.' The form is divided into two main columns. The left column, 'Personal Information', contains fields for Name (Test User), Age (25), Gender (Male), Height (cm) (170), Weight (kg) (70), Activity Level (Moderately Active (3-5 days/week)), Fitness Goal (Weight Loss), Diet Type (Omnivore), and Food Allergies (optional) (e.g. nuts, dairy, gluten). A 'Save Profile' button is at the bottom. The right column, 'Your Calorie Targets', displays calculated values: BMR (1643 kcal), TDEE (2547 kcal), and a 'DAILY CALORIE TARGET' (2047 kcal for Weight Loss). Below these are three goal-based targets: Protein (179g), Carbs (179g), and Fat (68g). A section titled 'About the Calculation' explains the formulas for BMR, TDEE, and Goal Adjustment (Weight Loss: TDEE - 500 kcal, Muscle Gain: TDEE + 300 kcal, Maintenance: TDEE (no change)). The footer shows '© 2025 Health+ Platform | All Rights Reserved'.

Health+

Home Dashboard Diet Plan Tracker Grocery Profile Reminders

My Profile

Set your personal details for accurate calorie and nutrition calculations.

Personal Information

Name
Test User

Age: 25 Gender: Male

Height (cm): 170 Weight (kg): 70

Activity Level: Moderately Active (3-5 days/week)

Fitness Goal: Weight Loss Diet Type: Omnivore

Food Allergies (optional): e.g. nuts, dairy, gluten

Save Profile

Your Calorie Targets

BMR: 1643 kcal

TDEE: 2547 kcal

DAILY CALORIE TARGET: 2047 kcal for Weight Loss

Protein: 179g Carbs: 179g Fat: 68g

About the Calculation

BMR (Basal Metabolic Rate) is calculated using the Mifflin-St Jeor equation — the most accurate formula for estimating your resting energy expenditure.

TDEE (Total Daily Energy Expenditure) multiplies your BMR by an activity factor to estimate how many calories you burn per day.

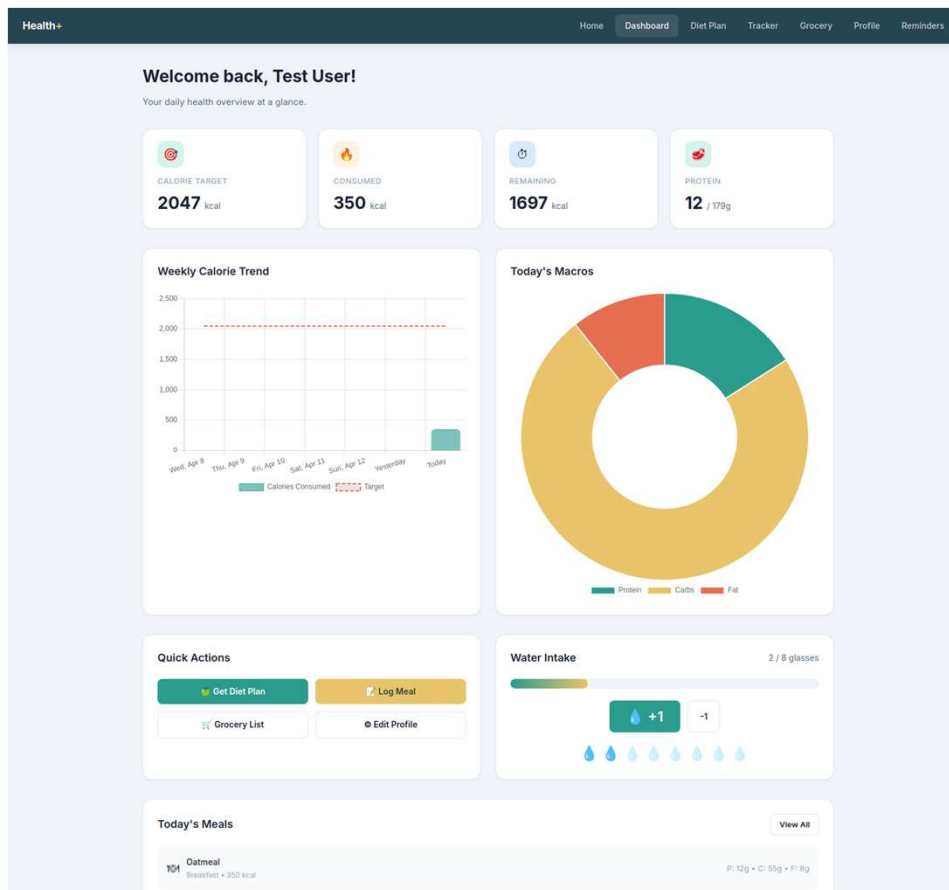
Goal Adjustment:

- Weight Loss: TDEE - 500 kcal
- Muscle Gain: TDEE + 300 kcal
- Maintenance: TDEE (no change)

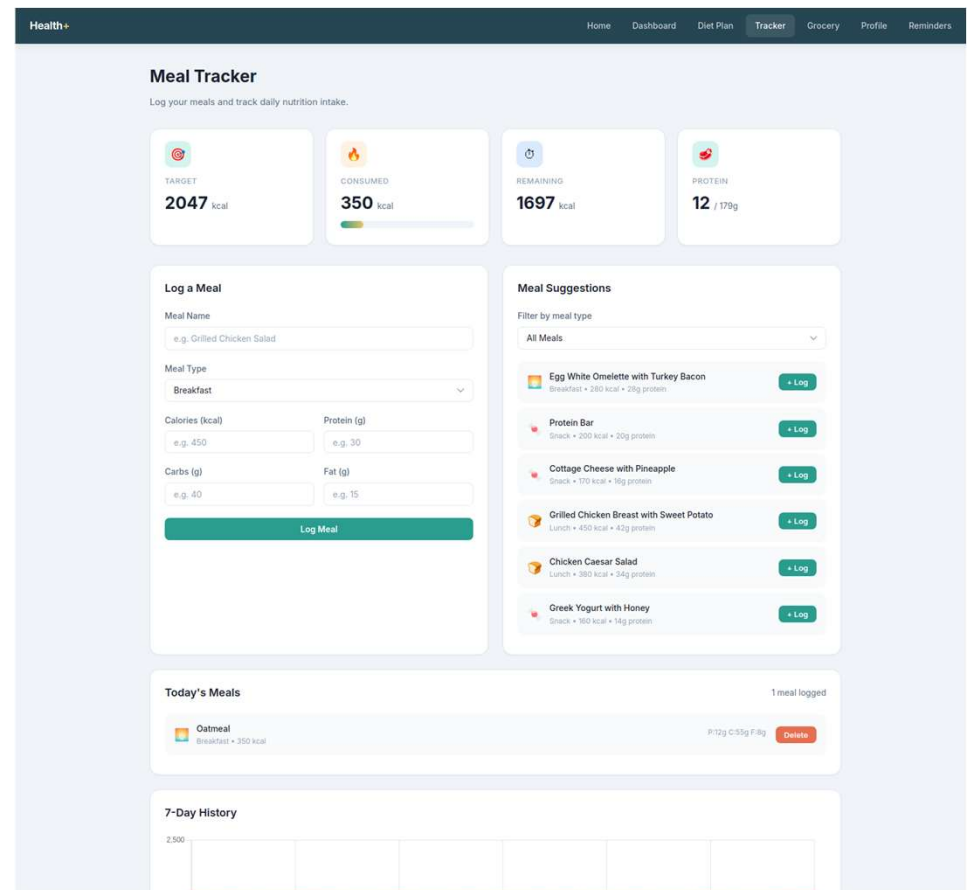
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7. UI/UX Design & Screenshots

Dashboard — Stat Cards, Charts, Water Tracker



Meal Tracker — Log Meals, Progress



7. More Screenshots

Grocery List — Auto-Generate & Manual Add

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HomeDashboardDiet PlanTrackerGroceryProfileReminders

Grocery List

Auto-generated from your diet plan or add items manually.

TOTAL ITEMS
0

CHECKED OFF
0

Shopping List

Clear CheckedClear All



Your grocery list is empty. Add items or generate from your diet plan.

Add Item

Category

Produce

Quantity

e.g. 2 lbs

Add to List

Auto-Generate from Diet Plan

Generate a weekly grocery list from your saved diet plan. This will extract all ingredients and organize them by category.

Generate Grocery List

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AI Diet Planner — Gemini Integration

Health+

HomeDashboardDiet PlanTrackerGroceryProfileReminders

AI Diet Planner

Get a personalized daily meal plan powered by AI, tailored to your body and goals.

YOUR DYNAMIC CALORIE TARGET
2047 kcal/day

BMR: 1643 • TDEE: 2547 • P: 179g C: 179g F: 68g

Your Details

Age

25

Gender

Male

Height (cm)

170

Weight (kg)

70

Activity Level

Moderately Active

Goal

Weight Loss

Diet Type

Omnivore

Allergies (optional)

e.g. nuts, dairy, gluten

Generate Diet Plan



Fill in your details and click "Generate Diet Plan" to receive a personalized meal plan.

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Testing & Security

Testing Techniques Used

Black Box Testing — Input/output validation without code review

White Box Testing — Code-level testing of calorie formulas

Integration Testing — Profile → Dashboard data propagation

Functional Testing — All 8 features verified end-to-end

UI/UX Testing — Cross-browser & mobile responsive checks

End-to-End Testing — Complete user journey validation

Test Results: 10 Unit Tests + 6 System Tests = ALL PASSED

Security Measures

API Key Protection — Server-side .env storage,
keys never exposed to client-side code

Input Sanitization — All form inputs validated
(Number(), max/min bounds, required fields)

CORS Configuration — Restricted cross-origin access

Error Handling — Graceful degradation on API failures

Data Privacy — All personal data stored locally in
browser (localStorage), never sent to server

Rate Limiting — API calls controlled to prevent abuse

No External Tracking — Zero analytics or cookies

Future Scope & Thank You

Future Enhancements

Cloud Database — Migrate to Firebase/MongoDB

User Authentication — Login/signup with OAuth 2.0

Progressive Web App (PWA) — Offline access + install

Barcode Scanner — Scan food items for nutrition data

Wearable Integration — Sync with fitness trackers

Social Features — Share meal plans with friends

ML-based Recommendations — Learn from user habits

Multi-language Support — Hindi, Gujarati, etc.

Thank You!

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